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Semester: IV
Subject: TOC

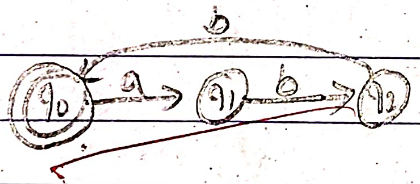
Q) Design a DFA with input alphabet $\Sigma = \{a, b\}$ and machine should accept 'a' followed immediately by "bb".

Solution,

$S_1 = \Sigma = \{a, b\}$

$S_2 = \min = a, b$

S_3:



Transition function

$\delta(q_0, a) = q_1$

$\delta(q_0, b) = q_0$

$\delta(q_1, a) = q_3$

$\delta(q_1, b) = q_2$

$\delta(q_2, b) = q_0$

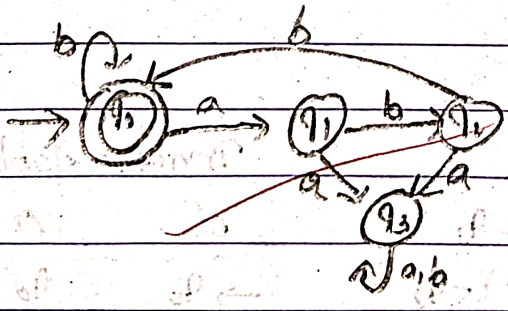
$\delta(q_2, a) = q_3$

$S_4 = \delta(q_0, b) = q_0$

$\delta(q_1, a) = q_3$

$\delta(q_2, a) = q_3$

S_5:



Transition table

δ	a	b
$\rightarrow q_0$	q_1	q_0
q_1	q_3	q_2
q_2	q_3	q_0
q_3	q_3	q_3

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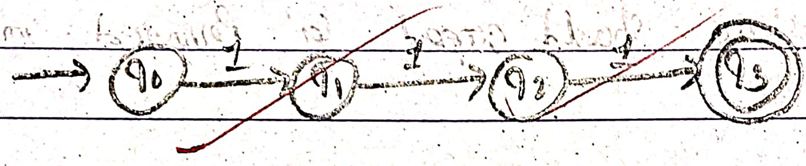
Q1) Design a DFA to accept strings having 111 as substring

Solution

S_1 = input alphabet, $\Sigma = \{0,1\}$

S_2 = min string = 111

S_3 =



S_4 = $\delta(q_0,0) = q_0$

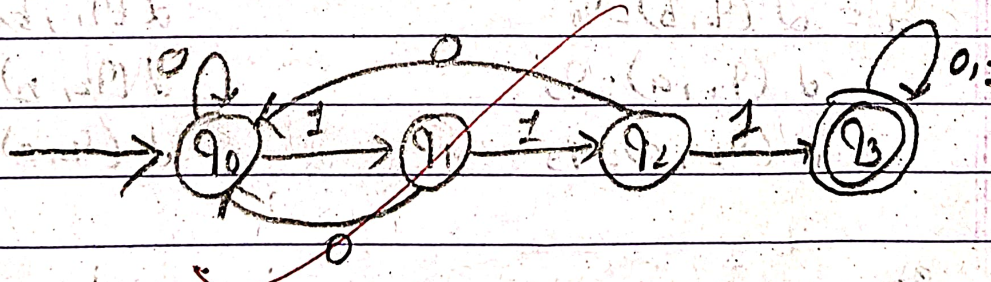
$\delta(q_1,0) = q_0$

$\delta(q_2,0) = q_0$

$\delta(q_3,1) = q_3$

$\delta(q_3,0) = q_3$

S_5 -



Transition functions:

$\delta(q_0,0) = q_0$, $\delta(q_0,1) = q_1$

$\delta(q_1,0) = q_0$, $\delta(q_1,1) = q_2$

$\delta(q_2,0) = q_0$, $\delta(q_2,1) = q_3$

$\delta(q_3,0) = q_3$, $\delta(q_3,1) = q_3$

Transition table

δ	0	1
$\rightarrow q_0$	q_0	q_1
q_1	q_0	q_2
q_2	q_0	q_3
q_3	q_3	q_3

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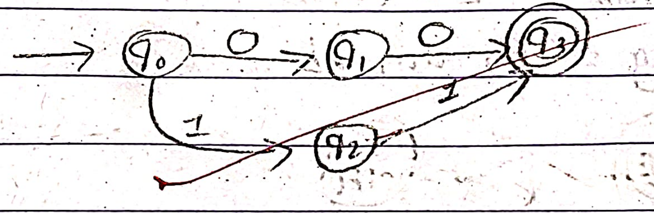
(2) Design a DFA that accepts strings that contains either two consecutive 0's or two consecutive 1's.

Soln,

S_1 = Input alphabet, $\Sigma = \{0,1\}$

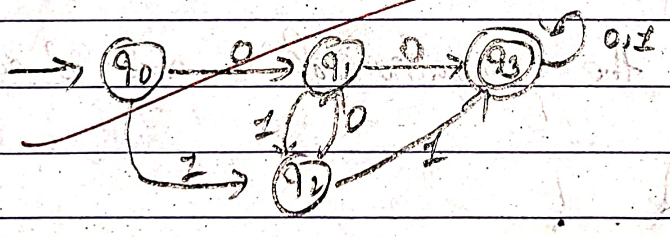
S_2 = min string = 00, 11

S_3 =



S_4 = $\delta(q_1, 1) = q_2$ $\delta(q_3, 0) = q_3$
$\delta(q_2, 0) = q_1$ $\delta(q_3, 1) = q_3$

S_5 :



Transition function:

$\delta(q_0, 0) = q_1$
$\delta(q_0, 1) = q_2$
$\delta(q_1, 0) = q_3$
$\delta(q_1, 1) = q_2$
$\delta(q_2, 0) = q_1$
$\delta(q_2, 1) = q_3$
$\delta(q_3, 0) = q_3$
$\delta(q_3, 1) = q_3$

Transition table

δ	0	1
$\rightarrow q_0$	q_1	q_2
q_1	q_3	q_2
q_2	q_1	q_3
q_3	q_3	q_3

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(3) Obtain a DFA to accept strings of a's and b's having even numbers of a's and b's.

Solution,

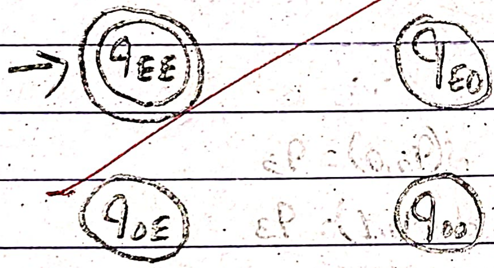
L ($n_a(w)$ mod 2 and $n_b(w)$ mod 2)

Total states = 4 $\therefore$ final = Even (a) & even (b).

S_1 = input alphabet, $\Sigma = \{a, b\}$

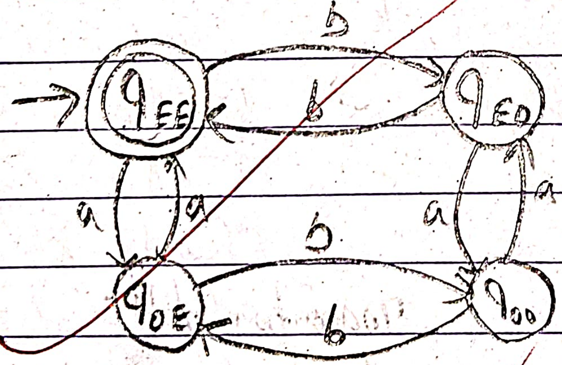
S_2 = min string = ϵ, aa, bb

S_3 =



S_4 = find all the transaction of 'a' and 'b' on each states

S_5 =



Transition function

$\delta(q_{EE}, a) = q_{OE}$ $\delta(q_{EE}, b) = q_{EO}$
$\delta(q_{OE}, a) = q_{EE}$ $\delta(q_{OE}, b) = q_{OO}$
$\delta(q_{OO}, a) = q_{EO}$ $\delta(q_{OO}, b) = q_{OE}$
$\delta(q_{EO}, a) = q_{OO}$ $\delta(q_{EO}, b) = q_{EE}$

Transition table

δ	a	b
$\rightarrow q_{EE}$	q_{OE}	q_{EO}
q_{EO}	q_{OO}	q_{EE}
q_{OO}	q_{EO}	q_{OE}
q_{OE}	q_{EE}	q_{OO}

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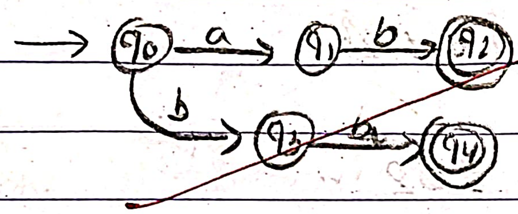
(5) Obtain a DFA to accept a language L that has strings of a's and b's ending with ab or ba.

Solution,

$S_1 = \Sigma = \{a, b\}$

$S_2 = min = ab$ or ba

S_3 :



S_4 : $\delta(q_1, a) = q_1$

$\delta(q_2, a) = q_4$

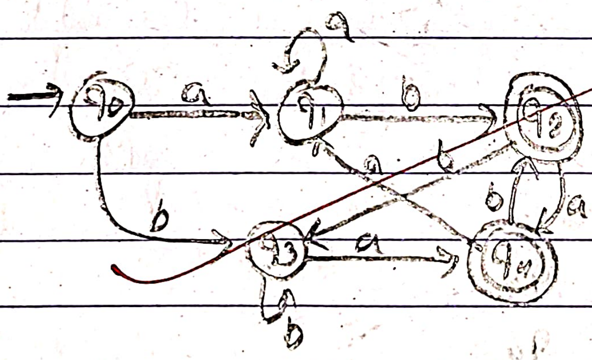
$\delta(q_2, b) = q_3$

$\delta(q_3, b) = q_3$

$\delta(q_4, a) = q_1$

$\delta(q_4, b) = q_2$

S_5 :



Transition function

$\delta(q_0, a) = q_1$

$\delta(q_0, b) = q_3$

$\delta(q_1, a) = q_1$

$\delta(q_1, b) = q_2$

$\delta(q_2, a) = q_4$

$\delta(q_2, b) = q_3$

$\delta(q_3, a) = q_4$

$\delta(q_3, b) = q_3$

$\delta(q_4, a) = q_1$

$\delta(q_4, b) = q_2$

Transition table

δ	a	b
$\rightarrow q_0$	q_1	q_3
q_1	q_1	q_2
q_2	q_4	q_3
q_3	q_4	q_3
q_4	q_1	q_2

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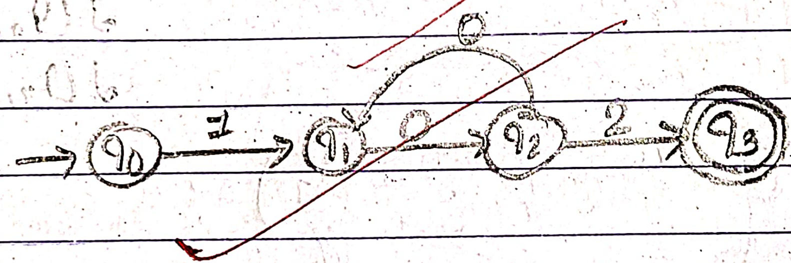
(4) obtain a DFA to accept strings of 0's, 1's and 2's, begining with a 1, followed by add number of 0's and ending with a 2?

Solution,

S_1 = input string, $\Sigma = \{0, 1, 2\}$

S_2 = min string = 102

S_3 =



S_4 = $\delta(q_0, 0) = q_4$

$\delta(q_0, 2) = q_4$

$\delta(q_1, 1) = q_4$

$\delta(q_1, 2) = q_4$

$\delta(q_2, 1) = q_4$

$\delta(q_3, 0) = q_4$

$\delta(q_3, 1) = q_4$

$\delta(q_3, 2) = q_4$

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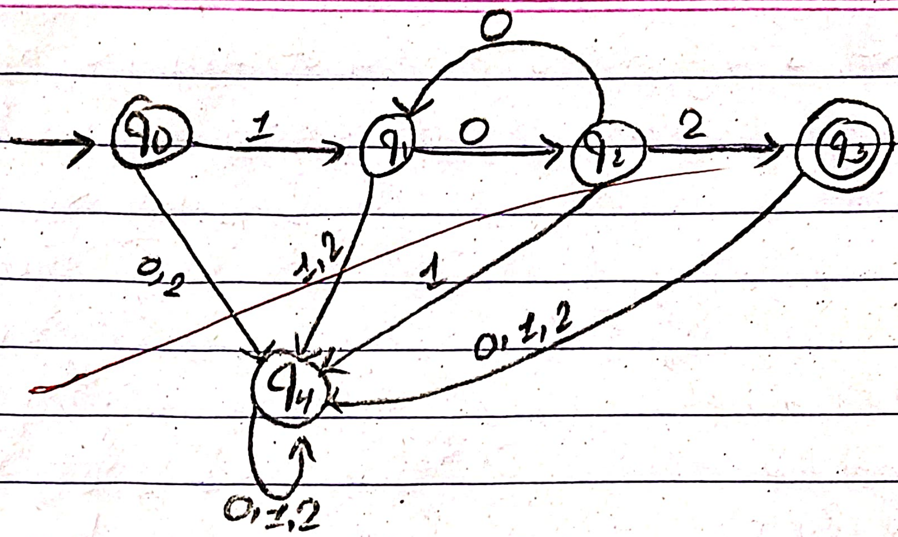
Ss:



Transition ~~table~~ function

$\delta(q_0, 0) = q_4$

$\delta(q_0, 1) = q_1$

$\delta(q_0, 2) = q_4$

$\delta(q_1, 0) = q_2$

$\delta(q_1, 1) = q_4$

$\delta(q_1, 2) = q_4$

$\delta(q_2, 0) = q_1$

$\delta(q_2, 1) = q_4$

$\delta(q_2, 2) = q_3$

$\delta(q_3, 0) = q_4$

$\delta(q_3, 1) = q_4$

$\delta(q_3, 2) = q_4$

$\delta(q_4, 0) = q_4$

$\delta(q_4, 1) = q_4$

$\delta(q_4, 2) = q_4$

Transition table

δ	0	1	2
$\rightarrow q_0$	q_4	q_1	q_4
q_1	q_2	q_4	q_4
q_2	q_1	q_4	q_3
q_3	q_4	q_4	q_4
q_4	q_4	q_4	q_4

A+

17/02/26